ASAP CRN Postmortem-Derived Brain Sequencing Collection

README - currently release 1.0.0-beta, 10.5281/zenodo.10372107

Overview

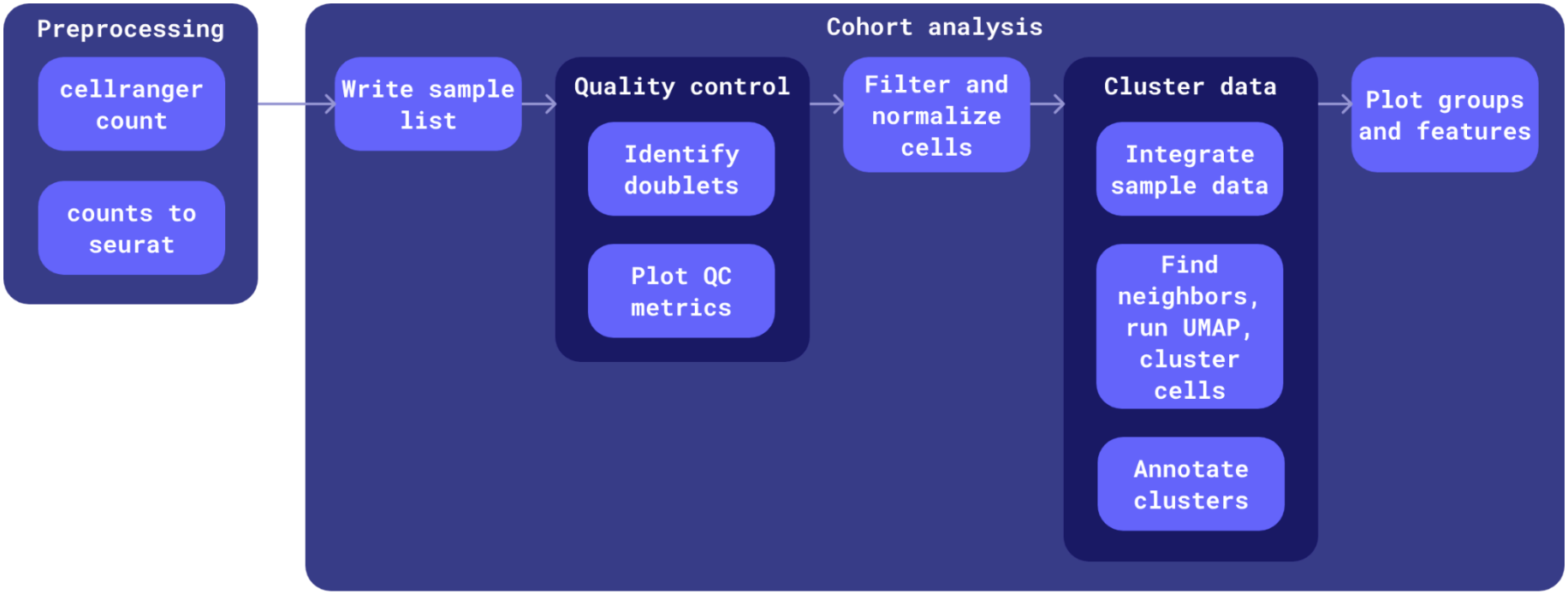
The Postmortem-derived Brain Sequencing Collection is a harmonized repository of scRNAseq sequencing data contributed by three ASAP CRN teams (Team Hafler, Team Lee, Team Jakobsson). These samples were derived from a variety of brain regions including middle frontal gyrus, hippocampus, substantia nigra, and pre-frontal cortex regions. They represent brains of healthy controls, Parkinson's Disease, and Alzheimer's Disease patients. The samples have been harmonized across 10x sequenced data aligned into count tables with cell ranger.

The current collection represents the *beta* release and will be expanded and improved as additional data is uploaded into the ASAP CRN Cloud. When complete, the collection will provide data generated from 1,800+ samples using proteomics, transcriptomics, and sequencing (single-nucleus RNAseq, single-cell RNAseq, bulk RNA-seq, ATAC-seq, long read WGS, and single-nucleus multi-ome sequencing (paired snRNAseq, snATACseq)) techniques.

ASAP Teams: Team Hafler, Team Jakobsson, Team Lee
Principal Investigator: Le Zhang, Molly Gale Hammell, Jose Bras
Contributors: Andy Henrie, Hampton Leonard, Heather Ward, Karen Fang, Sreeganga Chandra, Biqing Zhu, Haowei Wang, Anthony Russo, Jae Min Park, Celia Kun-Rodrigues, Rita Guerreiro, Le Zhang, Molly Gale Hammell, Anita Adami, Talitha Forcier, Raquel Garza, Annelies Quagebeur, Yogita Sharma, Oliver, H Tam, Cole Wunderlich, Roger Barker, Agnete Kirkeby, Johan Jakobsson, Jose Bras, Lee Marshall, Kimberly Paquette, Kaitlyn Westra, Andrew Pyman
GitHub: https://github.com/ASAP-CRN
Data Release Date: 1 April 2023
Release Number: 1.0.0-beta
Release memo: 10.5281/zenodo.10372107

Platformed Data

The raw scRNAseq data has been processed and harmonized to create a harmonized database of gene expression data. The platformed data is summarized below as Raw Data, Curated Data, and QC Summaries below. The complete workflow used for processing the platformed data can be found at the [ASAP CRN github repository](#).



Data File Manifest: ASAP CRN Cloud File Manifest - v1.1

Raw Data

The *raw* data refers to the fastq files transferred by each ASAP CRN Team. These data are available with the caveat that they are stored in "requester pays" buckets on the google cloud platform. A google cloud billing account will need to be registered to access these data. More information is available HERE.

Curated Data

The curated data can be categorized as "*preprocessed*" and "processed" , and "*integrated*".

Preprocessed. Pre-processing refers to the first stage of sequence alignment from the *raw* fastq files with cellranger for each sample, yielding count matrices (`*.filtered_feature_bc_matrix.h5s`) and an AnnData data object (`*adata_object.h5ad`) plus basic QC for each sample. This basic QC involves estimating the ambient RNA contamination, doublet detection, and identification of mitochondrial and ribosomal genes.

Processed. Processing refers to the next level of QC – filtering – and normalizing the count objects for each sample. The filtering and normalization of the data objects yields intermediate artifacts (e.g. `*merged_adata_object.h5ad`, `*merged_adata_object_filtered.h5ad`) and as per this workflow.

Integrated. The integrated or *harmonized* data represents batch-corrected aggregated data. Batch correction and integration of the individual samples into a single count table is the critical step of harmonization. In the workflow we call this entire process the "cohort analysis", and the key steps of integration take place in the "clustering" workflow.
The full *integration* is achieved through the following steps:

1. batch correction on the aggregate data with scVI method (e.g.`*adata_object.scvi_integrated.h5ad`), and
2. Clustered and annotated by creating embeddings, clustering on them, and annotation with CellAssign (e.g. `*adata_object.scvi_integrated.umap_cluster.h5ad`, `*adata_object.scvi_integrated.umap_cluster.annotate_cells.h5ad`)

QC Summaries (Plots)

As advertisements to the harmonized data a family of plots illustrating the Quality Control (QC) statistics and an overall illustration of the dataset. Embedding plots showcasing the distribution of batch labels, doublet scores, cell types, nCount, nFeature, percent-mitochondria, percent0-ribosome, and samples are generated. As are a scatter of UMI count vs Gene count, and violin plots for the preprocessing QC metrics (nCount_RNA, nFeature_RNA, pct.mt, pct.rb, doublet_scores)

Metadata & Data Dictionary

The ASAP CRN Common Data Elements (CDE) outlines the variables which are harmonized across the ASAP CRN. These metadata enable the harmonization of the data submitted by multiple teams. The metadata consists of five tables, which are described below, and also detailed as Data Dictionaries for each.

Data Dictionary: ASAP CRN Data Dictionary - v2

Metadata refers to descriptive information about the overall study, individual samples, all protocols, and references to processed and raw data file names. Metadata was aggregated using the table-based submission for each team's dataset. The tables - STUDY, PROTOCOL, SUBJECT, SAMPLE, CLINPATH, and DATA - can be thought of as spreadsheets, and each table is prepared as simple .csv files which constitute the tables outlined in the ASAP CRN CDE (final sheet). These have been aggregated across submissions and unique ASAP IDs generated for each dataset, team, subject and sample. I.e. ASAP_dataset_id, ASAP_team_id, ASAP_subject_id, and ASAP_sample_id.

The metadata can be browsed on ASAP CRN Cloud *Explorer* and exposed in the Verily workbench.

This document.